ActInf GuestStream 063.1 ~ "Incorporating (variational) free energy models

GuestStream | 1:41:25

right hello and welcome this is active inference live stream 63.1 it's november 7th 2023
we're here with michael pikarski and we'll be hearing a presentation followed by a discussion
on incorporating variational free energy models into mechanisms the case of predictive processing
under the free energy principle so looking forward to this lecture and discussion thank
you for joining and off to you thank you very much daniel thank you for invitation uh this is a great
honor for me and very nice uh i'm a little stressed i hope my presentation will be
interesting for you and give a chance to take a new some new ideas concerning the bayesian models
especially predictive processing under the free energy principle so my talk is based on my recent
papers which i published after after many many months of revisions in centers and i will sketch
the view which i try to develop here which i think has some interesting implication
also for the discussion in the field of bayesian modeling like in the wider context of philosophy of
neuroscience of philosophy of cognitive science so i will try to justify the free hypothesis
one general and two more specific concerning uh charlie the predict processing and free energy
principle so the first one idea which i found in a recent paper of some philosophers of mechanistic
explanations is there is the idea that there are phenomena i think that there are neurocognitive phenomena
which should be explained not only in the terms of the composing the mechanism which
uh is related to the this phenomena but also should incorporate the the constraints from the
environmental constraints and the flows of free energy i will be i will defend the view according to
which if you want to explain those mechanism those phenomena we have to take an account of flows of free
energy and the notion of free energy of course in the work of those mechanics uh is related to the
thermodynamic dynamic free energy but i will try to show that in the free energy we can understand
also in the terms of informational or variational free energy this is the first hypothesis the second
uh hypothesis is this is not my idea because we found in many papers of uh jacob howey of or pavel
gwarzyeski the idea that predictive processing provides a sketch of mechanism in other words predictive
processing is computational framework which has mechanistic or in which can offer a mechanistic
explanation of its target phenomena and i will try to to connect the two my hypothesis and shows that
mechanistic predictive processing takes a informational constraint from the free energy principle in other words
we can uh inc integrate uh free energy principle as normative theory normative principle with process theory
predictive processing using the mechanistic uh picture mechanistic idea of explanations so let's start from the
first hypothesis um i think that there is a general and uncommon agree uh in the world field in the works of
some authors from cognitive uh science from philosophy of neuroscience uh the view which
schematically called called new mechanical philosophy according to which the explanation of phenomena is a
matter of the composing this phenomena into underlying mechanism so when we want to explain phenomena for

example this is a view to defend the the car craver or william bechtel and the others when we want to explain

phenomena not only cognitive but also biological social phenomena we have to decompose the mechanism which

is responsible for the realization of this phenomenon and this mechanism can be treated as a

uh mechanism which casual product the phenomena or a mechanism which is responsible in the terms of constitutive

responsibility for the realization of the phenomena and uh those authors who and those

start these traditions which uh craver in his book explaining the brain calls the system traditions claims that

explanation is a matter of the system that explanation is a matter of finding the systems the mechanism which

should be described should be should be identified by scientists and should be decomposed into into parts

into components we should recognize the relation between those components and the we should recognize and describe the organization

which rules of the relations between the components and it means that we can think about the mechanism in terms of

hierarchical organization because if we look on this figure from the book of the craver phenomena phenomena are based

hierarchically on the underlying uh structure of hierarchical structure of mechanism the lower and higher level mechanism and according to our

uh scientific aims according to uh our strategy we can't uh looking for the more deeper or uh more deeper mechanism concerning for example uh how would

the level of biological organization to the level of the physiological organization and so on and one of the

uh most important authors in the field of mechanistic explanation the field of uh new mechanism william bechtel um in recent latest paper uh

he little change the standard view on the composition he claims that heuristic of the composition or normative strategy of the composition mechanical

uh explanation should be um um

um

rethinking we should rethink this strategy why because there are

um high-level cognitive mechanisms

which cannot be described solely in the terms of hierarchical

organization or in the terms of casual

uh relations because

the conditions because there are some kind of mechanism which are as

uh bechtel and colleagues claims

which are active control mechanisms

which are active control mechanisms

the mechanisms which are responsible for production

the phenomena

what kind of phenomena they are active phenomena active active

structures like uh
bodily movements responsible for body movements and physiological
process and craver and bechtel claims sorry
that those
mechanisms
of course they play a central uh
uh casual role
in the hierarchical organization
but they also a components
of complex network of heterological
web of control systems and
they play its casual role
their casual role because
they perform a given function
in this heterological web
the hierarchical web
why it is possible because
in this sense though
those phenomena are the
are mutually constrained by the other
elements
we can call their constraints
and they are
and they are active
or they derive their casual
efficiency because they are constrained
systems and
in the opinion of
bechtel
and
the other authors who co-work with him
it is important to rethink
uh our old
image of the composition
because
the standard view on the decomposition
we don't take an account
of the

two elements on the
constraints
which make given
mechanism
active in the sense of
the being part of a heterological web
the second point is the energy
because biological and
mechanisms cognitive mechanism
are always
um related to the
flows of the free energy which they
uh transform which they
uh process
and
why this is important because according
to bechtel he claims
what is characteristic for biological
mechanism
in the context in the contrast to the other
mechanics is that the biological mechanism
are dissipative structures
it means that they occur and maintained
in the context in the fridge in fridge free energy is being dissipated
and
if we
don't
incorporate
this element of transforming of
uh processing the flows of free energy
we
in the fact that we don't really explain why
given mechanisms or given systems
have behaviors
as bechtel and biff claims
completely unconstrained system
will have no behaviors
and

i will think
that
um
if we want to explain
neurocognitive mechanism we have to
adopt the view which i described in my
paper as a constraint based mechanism
approach and according to view
it's heuristic mechanistic explanation
of at least some
cognitive phenomena
should be based or should incorporate
dimensions of constraints
element of the environment
which makes
mechanism active
and the flows of free energy
and this is
the first piece of
uh my idea and
second is that
i think that we can try to
demonstrate this idea in
uh
one
example from cognitive
contemporary cognitive
science and i will try to
adopt this idea into the frame
of predictive processing so
now i will sketch
a little a short view
about predictive processing framework
and why this is important for me
uh to try to test this this framework
into the context of the constraint based
mechanism approach
um so predictive coders

uh develop the um
country and the helmsholzian idea that
that our cognitive systems our brains
don't have indirect access to the causes of information we receive by the input
and the uh and they claim
that uh the central the main function
of the brain on the search and nervous system
is to minimize uh uncertainty which is described in the terms of minimizing long term minimization of average prediction error
uh how it is possible how brain can do this uh it is possible because the brain inhabit or embedded the virtual informational structure which predictive models by
uh described as generative
and the generative model is hierarchical
and multi-level
structures which transform
prog- process the information coming from the intus
why because in this way
the brain can minimize the
so-called prediction error
the discrepancies
the differences between
the state of the model
state of the organism
and the information coming from the
from the environment
because successful
minimalization of prediction error has
many adaptive
uh benefits
so uh this is
general uh picture
now we should look
more uh closer
to this image and try to
develop a more
formal technical language
so
uh we can

use this

all object
to think about

relation between the events in the world

term of
relation between cause
and effect

what we receive is the observation
we receive by the
sensory input observations
and if the idea
developed by predictive coders
coming from helmsholtz is that
we don't have an access to the straight
true states
as understood as the causes of the observation
so we have to infer the true states
to act successfully in the environment
and how it is possible to infer
to guess the true states
responsible for the observation we receive
so there is a development of the
helmsholtzian idea of
unconscious inference
and bayesian claims that
this idea can be translated into the
thinking about the statistical inference
what kind of statistical inference
as we know this is the bayesian
bayesian inference but
what does it mean the generative model minimize prediction error or infer the true states
using the bayesian inference
talking about bayesian inference

rises
a small problem because bayesian inference
exact bayesian inference is
incredibly hard
you cannot compute exact bayesian inference
and the idea which developed
for example carfreestone is that
our models
our brains using models
generative models
don't compute exact bayesian inference
but it uses the simpler version
of bayesian inference
they approximate
they use approximation of bayesian inference
so in other words
if the story is correct
then
continuous minimization
of prediction error
can be described or understood
or explained in the terms of approximation
of bayesian inference
and this approximation is possible
because model
minimizing prediction error
simultaneously minimize
a certain quantity
which freestone calls
the quantity which
is always greater than
to suppressor

this quantity
model
iteratively update
the internal parameters
update the internal states

and
in that way
can infer
the most probable
true states
which is responsible for the observation
which model
refers
so
according to predictive
co-processing
as I read it correctly
what is characteristic
for the brain
is it that the brain
main function
is to
processing information
in the terms of bayesian approximation
and I try to unpack this idea
and
read this
and try to read this
in the context of
new mechanical philosophy
so
this is the first step
the second one
is to justify
that predictive processing
can be
connected
with mechanical
philosophy
and
in many papers
of
Kovává G?adziecki

Harkness
Kshava
Howey
we
we see good arguments
according to which predictive processing
as computational architecture
provides
the scientist
the so-called sketch
of mechanisms
what is sketch of mechanisms?
what is sketch of mechanisms?
this is the idea
developed by
Piccinini and Kaplan
and they claim that
sketch of the mechanism
is the incomplete
representation of mechanism
it means this is
some kind of representation
of mechanism
when the scientist
omitted
some components
some parts
of the mechanism
and good sketch of the mechanism
should lead us
to the schema of mechanism
which is the complete representation
of constitutive element
of given mechanism
so
if predictive processing
give us a chance
to write a sketch

of the mechanism
and
if predictive processing
is about
the neural function
of the brain
then
if the view
which I developed
after the ideas
of Bechtel and colleagues
if this view
of the constraint-based
mechanism approach
is correct
then
I will claim
that
if predictive processing
if predictive coders
want to
build
a mechanistic explanation
of cognitive phenomena
they should include
into the explanation
flows of
free energy
because
without
this
flows
without the constraint
of information
or
in general
free energy
we cannot

explain
how it is possible
that mechanism
operates
and is active
as long as
this energy is available
and the thesis
which I defend
in this paper
in sentence
is that
predictive processing models
should include
informational constraints
if they are to be mechanistic
of course
if we
think about
predictive processing
as the way

the function
of the brain
and
of course
there is
one
natural question
because many authors
will claim that
the good
theoretical
normative
principle
for building models
in predictive processing
offer the

free energy principle
so
the question
arise

the free energy principle
can offer
these constraints
for predictive
architecture
and I will be
claim that

the FAP
offers
these constraints
but
of course
we should
try to
justify
this
this thesis
more
specifically
so
but
we've got
we find
the one
big problem
when we
look into
the
2018
paper

William Bechtel

when he
introduced
the idea
of
the necessity
of
including
the free energy
flows
into explanations
he
directly
writes
that
the notion
of free energy
which he developed
in his papers
in his view
is distinct
from the
free energy
principle
or free energy
articulated
by Carl Friston
and the others
in other words

that

we should include
into explanations
flows of the free energy
but this is the thermodynamic
!
free energy
we cannot use

the other kind
of the energy
because
because this is distinct notions
in the same year
in 2018
that
Freiston
published
a draft
or a preprint
of his book
Free Energy
Principle for Particular Physic
and he said something
that contradicts
with the view
of the Bechtel
he said that
heuristically
free energy
in the term
in the term
in the term
in the term
in the term
in the term
in the term
in the term
that Freiston claims
something
that contradicts
the thesis
of the Bechtel
and I think
that it is possible
to justify
the thesis

of Freiston
but possibly
I try to do this
in a different way
than
than Freiston
does this
so
for me
as a person
without good
formal
preparation
to reading
the paper
of
Carfriston
it was hard
to
find a good way
of reading
his paper
and his cooperators
but I think
that we can try
to reconstruct
the idea
which supporters
of the free energy principle
develop
and these ideas
can be commonly
!
described
as something
what I call
the Bayesian mechanics
which supporters

of the free energy principle
develop
and these ideas
can be commonly
described
as something

the Bayesian mechanics
which

and
offer
the
theoretical
background
of the

argument
the first
side
I wrote
here is coming
from the paper
of Max Ramstad
and he claims
that
at the core
of Bayesian mechanics
is the
variational free energy principle
so that Freiston
in his latest paper
proposed
the idea
of the loping
of the Bayesian mechanics
which

can investigate
describe
the theoretical background
for explanation
what is free
energy principle
what is variational free energy
and
why this idea
is important
for me
because
Freiston will claim
that every kind
of mechanism
each mechanism
can be treated
as an expression
of some kind
of mechanics
we've got
classical mechanics
we've got
quantum mechanics
thermodynamics
statistical
also Bayesian mechanics
and every kind
of mechanics
has its own
reified constructs
and those constructs
those constructs
can be understood
as thermodynamic
energy
temperature
or variational

free energy
and
what is important
is that
that those constructs

reified constructs
those notions
are justified
and can be used
only in the realm
of the given
type of mechanics
it means that
it is possible
that we've got
some reified constructs
which we can use
on the base
in the realm
of the one kind
of the mechanics
but we cannot use
this in the other
type of mechanics
for example
when we talk about
the notion
of the
temperature
it can be
it will be
used in different
way
in the realm

and in the realm
of statistical

mechanics
and it means
that if we take
the construct
like variational
free energy
or thermodynamic
free energy
from the point
of the view
of this argument
it means that
all of these constructs
are relativized
or related
to the description
and method
of measurement
which we use
in the specific
type of mechanics
and what does it mean?
It means that
there is no
ontological
we don't have
a good ontological
argument
to claim something like that
that for example
statistical mechanics
is more primary
than for example
plastic
or
Bayezza mechanics

Because

every kind
of mechanism
each type of mechanics
can be
treated as a
complementary description
of the behaviour
of the dynamic systems
and it means that
the most primary
the most basic

we try to explain
and describe
in terms of
mechanics
is dynamical
systems
or the dynamical systems
the other
of the
mechanics

Freestone
it means that
Bayezza mechanics
can be treated
as all other
kind of mechanics
which we find
in physics
but what is
specific for this
mechanic as I said
is the assumption
of variational
free energy
which is strictly

related to the

Markov blankets
the constraints
of Markov blankets
Freestone will claim
that
this
additional constraints
the constraints
of Markov blankets
makes
possible
to speak
of states
of something
as relative
as something different
of something else
and why this is important
to make a possibility
to think
to talk about
for example
internal and external
states
because
this
talking
this
the possibility
of talking about
different states
directly applicable
and directly important
to the living organism
and for example
neural structures

Freestone in his book
claimed that
every
every type
of mechanics
you can ignore
assumption of
a Markov blanket
and the assumption

free energy
and
and
in some
implicit sense
will claim
that

outside blanket
and the states
outside blanket
this difference

kind of states
can be ignored
why
because
if this
way of thinking
is correct
every kind
of mechanics
give us a chance
give us a possibility
to talk about
the systems
we try to describe
in terms of

heat bed
or thermal reservoir
it means
that every kind
of mechanics
except
quantum mechanics
give us a possibility
to talk
about the states
as
in equilibrium
steady state
so
and according to
this argument
we should say

variational free
energy
can be applied
and justified
only in the realm
of Bayesian
mechanics
and this
application
help us
to describe
explain
why some
things
are autonomous
or active
why some things
can be as
or say
dissipative

structures
but
it means
that variational
free energy
give us a chance
to describe
some kind of
phenomena
some kind of
system
as autonomous
or active

and
simultaneously
thermodynamical
free energy
can be applied
in the realm
of the object
which we
can try
which we want
to explain
in the terms
of statistics
and this
is
this is
why
Freestone
will claim
that

variational
free energy
and thermodynamical

are consistent
because
they are two
consequences
or expression
we can say
the same
more elemental
thing
the thing
which can
more elemental
more basic
mechanistic
or maybe
quantum nature
dynamical
free energy
and variational
free energy
are the
notion
or constructs
which we can
trade

side
of the one
coin
and
using them
is dependent
of our
method of
measurements
of our
scientific

interest
from
our
strategy
to investigate
phenomena
they are

in the sense
that they
offer
the
scientists
the
possibility
of describe
dynamical

from the
two other

but those

are complementary
and how
we should
understand
and interpretate
the Bayesian
mechanics argument
because I think
this argument
should be interpreted
in a more
philosophical
more specific
way
if

the
thermodynamic
free energy
and variational

are two sides
or are related
to us
two aspects
of more
primal dynamics
there is still
open the questions
about what we talk
when we talk about
systems which minimize
variation of energy
and when we talk about
the direction
the flows of the
thermodynamic
of free energy
and I think that
this ultra realistic
or super realistic
interpretation
of this argument
is not
correct
because it seems
that we should
interpretate
a construct like
like
thermodynamic
and variational
free energy
as

useful
useful
functions
which
are
which
justification
are related
to our
scientific models
and so on
and in other
ways
there will be
the problem
to defend
the
thesis
that
the
Pyersian
and stochastic
mechanics
are equivalent
formulation of the same thing
I think the same thing

which
is the dynamical
systems
and
we try to
find
a good tools
to describe the
systems
but it means
that we don't have

a good tools
which represent
the
real structure
of the systems
and it means

we use
like
rational free energy
and thermodynamic

should be understood
as the constructs
which are related
and
ultimately reduced
to the
method
of measurement
descriptions
the method
we use
is

address the casual structure of the world in the sense that they are epistemically useful.

We can use them in the practice of model building, but they are not epistemically useful in the sense that we describe the real properties,

which we can literally describe as generative models, as variational free energy, and so on and so on.

And I think when we talk about the free energy principle, the instrumental and realistic interpretation,

is correct, the free energy principle doesn't imply any ontological solution about the target phenomena.

But from the point of view of my argument, there is one problem,

because this instrumental interpretation of the Freestonian argument

doesn't have any specific mechanistic implication.

We don't anything talk about mechanistic character of the phenomena.

Why this is important? Because when we come back to the field of new mechanical philosophy,

we all see that I think there is in very general, there is a common view that we should think about mechanism in a realistic manner.

What does it mean? The view which I try to describe here is so-called mechanistic realism,

according to which we should think about the structures, the entities in the world,

as the structures which are in some sense richer than mere aggregation of causes.

Why this is important? Because mechanism which we try to identify and decompose should be produced by some kind of structures.

And these structures should be more richer than a simple aggregation of the causes.

And this is the first part of this argument. And the second connected to the idea I try to develop is that there are some of those structures,

whose at first organization cannot be reduced to an aggregation of causes,

and second disorganization should be explained in terms of mechanisms which are constrained by flows of information,

flows of thermodynamic free energy, and described maybe in terms of minimizing rational free energy.

In other words, I will be claimed that if we adopt the mechanistic realism,

we should try to say that there are such phenomena, such structures,

the explanation of which should take into account the energetic constraint,

which we can try to describe in terms of variational free energy.

And if we try to adopt the image coming from the mechanistic realism,

we will defend the view which I describe as moderate realism about Bayesian modeling

or moderate realism about predictive processing and the free energy principle.

In this moderate realistic interpretations, we can say that system minimizes prediction error

or minimizes variational free energy because it implements in some way some casual mechanism

that can be described approximately in terms of minimizing variational free energy

or simultaneously maximizing mutual information between internal states and sensory states.

And it means that we can identify, describe, and decompose some kind of mechanism,

which we should trade as a system of mutual constraints that restrict the flow of information,

the flows also of thermodynamic free energy to perform work,

in such a way that those mechanisms minimize or should minimize discrepancy between internal state,

between the parameter of the model, between the prediction of the model,

and the information coming from the environment in which given systems act and live.

Why? Because without taking those constraints, flows of information,

we cannot describe and explain why those kinds of mechanisms or why this kind of organism

are at non-equilibrium steady state.

So, and I think that we can find in the field of thermodynamics of information and also in the work of Kraft-Riston,

argument which I called Argument from Neurocomputation,

a good argument for the moderate realistic interpretation of the PPP and the free energy principle.

And according to this argument, we can say that there is always trade-off between neural information processing

and thermodynamic energy cost.

It means that every time when biological sensors detect a change in the environment,

every time when biological organs process the information,

this processing information or detection change in the environment is proportional to the amount of information,

it's proportional to some minimum energy in terms of thermodynamics.

So, every time when systems process information,

every time there is minimum cost of thermodynamic energy.

And this argument from neural computation could be just related to the formal investigation

coming from the Fristow monograph about the Jerzy?ski Equality.

I think this piece of this book about the Jerzy?ski Equality is really hard.

But one and a half years ago, I was talking with Karl Fristow about this argument about the Jerzy?ski Equality,

and I think we can try to reformulate it in a more non-formal way.

And according to the Jerzy?ski Equality,

if we interpretate this in the realm of the free energy principle,

this equality allows us to connect variational free energy with thermodynamic free energy

in this sense that when we want to describe a moving one system from one state to another,

and this moving in the system we can understand as creating or destroying the information,

there is a certain amount of thermodynamic work cost and thermodynamic action which should be entailed by this moving.

In other words, when we look into the generative model in our brains,

we will say that every time when the model updates belief,

in the sense that every time when model changes information in the internal state of the model,

every time there is a thermodynamic work cost.

So the belief updating in the Bayesian generative model is strictly related to the thermodynamic cost of this belief updating.

And I think that moderate realistic interpretation should explain us how it's possible,

because when we talk about the belief updating generative model,

when we talk about the minimization of the error,

literally we don't map the formal structures we use onto the target phenomena.

I don't claim that there are real generative models in our brains,

I don't think that our brains optimize Bayesian and so on,

but I will claim that there are some structures in our head,

there are some structures in the wand that are richer than simple aggregation of cause,

and those structures implement some casual mechanism that scientists can approximately describe

in terms of models which minimize operational energy or prediction error.

And I agree with the thesis of Kirchhoff and colleagues that our models

fits the data without literal mapping.

Yes, our models are true because they are approximation of the data.

And I think that we can think about the predictive processing model incorporated or related,

or integrated, sorry, with the free energy principle as some kind of approximation of the data which we want to investigate, a specific type of the data, like our neural organs and so on.

And I think that we should distinguish and remember about three distinct elements.

The free energy principle as formal principle, the formal principle which doesn't assume any ontological assumption

and doesn't imply any ontological commitments about the target phenomena.

the predictive processing framework as mechanistic computational models,

which are grounded from the one side in this formal principle,

which the formal principle offers the variational methods, variational notions,

and from the other side on the heuristic of mechanistic explanations.

And the first, the biological systems, the target systems that predictive processing and biologians employed to model,

which are based, which are independent of the implication or on the language of the free energy principle,

which are independent from the language and ontological implication of process theory like predictive processing.

So, and going to conclusion.

So, I will claim that the reference to the informational constraints

allows the scientists to explain the neural organs, the brains,

not only as a kind of physical material, physical mechanism,

but also as a systems which have a future homoesthetic nature.

So, and I will claim that it means that a full satisfaction or explanation of how the brain works,

not only how the brain can be treated in terms of statistical or classical mechanics,

but also in terms of thinking about the brain as active as part of living body,

a full explanation of brain as part of the living body requires taking into account informational constraints,

which I believe can be characterized and described in terms of Bayesian operational free energy minimization.

And the last one conclusion is the observation that from the point of the view of the argument I tried to develop

is that the free energy principle is also normative in the sense that it sets a norm

that should be met by mechanistically non-trivial process theory or predictive processing models,

assuming the correctness of the constraint-based mechanism approach.

Okay. Thank you very much. That's all what I have.

Thank you. Great presentation.

I'll make a first comment.

Meanwhile, anyone who's watching live is welcome to ask a question.

Okay.

Well, I would like to applaud this great talk.

You connected the scholarship and the framings that are used in the philosophy of mechanistic accounts

with free energy principle, active inference, predictive processing.

And I think you articulated several lines of arguments that were in the literature, including where there was change or even partial incoherence or just partial ambiguity,

the fact that there was a lot of ambiguity, certainly ambiguity,

related to justifications or understandings of the relationship between

thermodynamic free energy and mechanics and Bayesian free energy and mechanics.

And that is a very wide space with everyone saying things like it's a purely formal resemblance.

And it's just a convenient computational heuristic used in science and physics and engineering.

And there's just no need to worry about more than that.

Ranging to arguments that they are co-extant or that they have different kinds of statuses.

So I very much appreciate in your paper and in this presentation that according to the way that different discussions about mechanism have been carried out in the biological sciences,

using that setup to clarify some of these discussions in this field that are all great to approach with kind of a first principles view.

And it's helpful to see how it connects with the arguments of like Craver and Bechdel.

Thanks. I agree that I try to develop some arguments which are, which exist in the current literature.

But for example, when I read the paper of Bechdel, I think that this idea

doesn't be directly or explicitly articulated.

And I think that Bechdel doesn't agree with my argument, my interpretation of his view, because of the variational free energy.

But I still believe that this is very, very basic intuition.

But this intuition is said that if we think about the part of the human organism, if we think about the brains,

what is more specific for the brains is that the brain processes information,

not brain processes the thermodynamic energy, not brain processes or produce some kind of hormones,

but the brain processes the information. This information is very important for a living organism.

And I think that this intuition should be tasted in the field of science or maybe in the field of philosophy of neuroscience.

And I'm not so sure that this is possible to taste it, but I think that this is important to the description or explanation of those mechanisms,

that those mechanisms are the mechanisms of transforming, processing information.

And possibly, possibly, possibly, maybe it's enough to say that, okay, those mechanisms of transformation with the brain,

which the brain processes, is not important to explanation of how it is possible.

But I think that predictive coders, the supporters of the free energy principle, said something important.

And they claim that what is specific for the brain is not only that the process information, but also is the process in some specific way,

using something what we can describe in terms of Bayesian approximation or more wider as a statistical inference.

And I think that if predictive coders, predictive processing is a fruitful strategy of explanation,

in the field of neuroscience, if the model builds by predictive coders are scientifically useful, then we should use this heuristic coming from the free energy principle, that take it seriously, because I don't want to, I try to say that,

talking about the brain which uses, which uses, which uses, which uses the generative model which approximates Bayesian inference,

this is not the way of talking, way of speaking.

I think that we not only describe, but the word we use, the notion we use, as I said, in some way corresponds to the,

some kind of structures, casual structures which we have in our head.

And maybe we investigate, we invite more specific, better words, better concepts, better tools to describe those structures.

But I believe that for now, predictive processing under the free energy principle gives us a good chance to explain why our brains are not only the matter,

but also the important part, but also the important part of the living bodies, and what is the difference between the brain as active control mechanism,

and why this brain shouldn't be treated as heat bath or thermal reservoir, and so on.

And I think that, and I think that, and I think this is the way, this is the reason why Bechtel tried to reformulate this,

the classical image of the composition, because he observes that some mechanism cannot be described or treated as non-living, non-active.

And if they shouldn't be treated as active, they should be treated as active and living, then I think we have to, we have to try to incorporate into our understanding,

into our explanation, the fact that the information which the brain is those mechanisms process,

is really, really important to understand why and how they act, how they are the part of dissipative structures like our organ.

Thank you. I'll ask a question next from the live chat.

John Athos wrote,

John Athos wrote,

What are you defining as information, when you say the brain is processing information?

Good questions.

I think, I think when I talk about information, I think about some kind of semantic information.

Of course, if we try to use in practice this idea, which I try to develop, the question about information should be more specific,

more information should be more specified. But I find the argument which developed Rosa Kao, as I correctly remember,

when she claims that when we talk about the information which is processed by single neuron, of course, this is some kind of syntactic information.

There is no semantic information. There is no semantic information. But when we want to try to understand how it is possible that the brain,

the brain, the organ, the cortex, process information, this is always strictly related to some kind of semantic information.

And I think we can think about semantic information.

Try to, but this is, I don't know for now how it is possible in the realm of my argument, but we think to try about this, what we traditionally describe as semantic information, in terms of variational information.

Because I think that if brains minimize prediction error, if brains approximate Bayesian inference, it means that brain operates, uses the information in semantic sense.

And in my paper, in my paper, I wrote about the mutual information between the internal and sensory states.

But I think, I see this as a lack of my view, because I don't precise what kind of information precisely I mean.

But I think that if we take this view as some kind of proposition of heuristic of building the models, then empirical investigations lead us to the possibility of describing this information in more specific terms.

Awesome. Very interesting. So a few lines that I'd love to ask questions about.

So in your paper and presentation, you center the variational free energy as kind of the analytical construct doing the work in the mechanistic account.

And I wondered what leads to one analytical construct doing that kind of work.

For example, why not center surprise and see variational free energy as a computational heuristic or just an approximator?

So what would be the relevance in a mechanistic account that centers the concept of surprise or the concept of prediction error versus the concept of variational free energy, which are related concepts in that an organism minimizing one does minimize across all.

So what is this is known to be

that notion we use, the concept we use, are related,

are corresponds to the structures which we try to develop

and describe using these notions,

but not the sense that we literally map the content of these notions

onto those target phenomena.

So probably varational free energy for now looks a good fit

to those phenomena from my point of view,

but it doesn't mean that we don't use the other notions.

But you ask why we don't talk about surprise or predictive errors.

I think that we don't explain in a satisfactory way

this idea when we talk about the surprise, maybe,

or prediction error because we don't understand

why brain approximates Bayesian inference.

If we don't introduce the notion of the varational free energy,

or maybe it's equivalent to the upper bound of surprise,
but if we don't introduce the notion,
then we don't find the computational formulation
or computational foundation
to description of the brain working
in terms of minimization of prediction error.
Because I've got some problems with it,
but maybe try to explain how I understand this.
If we take the fuel level of description of David Maher,
I think that talking about the minimization of prediction error
or minimization of uncertainty
are related to the computational level.
And I think that talking about minimization of varational free energy
or approximation Bayesian inference
are related to the algorithmic level.
And the idea which I try to develop
is using the connection between those levels
to investigate what's happened on the mechanistic level,
the different level of Maher.
And I think that the notion of prediction error or suprizo,
when we talk about this, we omit something important,
what describes and explain Friston in his paper
concerning the varational free energy.
Yeah, very interesting.
Also, there's different ways to view this,
but in the VFE centrism,
there's also the possibility of relationality
as a core principle.
Because if the mechanism surrounded surprisal of the system,
that would be closer to an account
where there was like a surprise module
or something that was about
what the systems,
what representations were internal for the system.
But with VFE,
we have something that's explicitly implementational
relative to a constructed model.
So then it naturally connects to discussions

about how modeling is done as a practice
because it is a constructed variable,
where as like temperature, ostensibly,
it is important for a chemical reaction.
Or temperatures of course,
graining, I'm sure there's a lot of ways to say it more formally,
but the kinetics do matter for the chemical reaction.
And then we have tools that are able to measure temperature
in different settings.
Cognitive modeling seems to have a relational step
that's a lot more multifaceted
because VFE is only calculated relative to the modeler's construct.
So
one person might make a model that shows the VFE doing this
and a different model would show something different.
and that seems to be a bit different than
like we have two thermometers,
but we both agree that temperature really does matter.
Yes, yes, I agree.
I agree.
But I think the information,
that the dimension of information is more subtle
than the dimension of temperature of mass.
And of course, VFE is related to the modelers
and model which they develop.
But
I try to sense something more.
Of course, VFE is related to the modelers,
but there is something in the target phenomena
which motivate modelers to use the VFE.
of course, if I look
on the piece of the paper,
the paper doesn't look like,
for example,
process the information.
And it doesn't motivate me to think about this paper
as something that I can model
and using the variation of energy or the VFE principle.

But when I talk about the living organism or the human brain
on something like that,
there is something in the structure,
in the phenomena which motivate me to build a model
more than using this principle.
And of course,
the truth of the method of measurement,
the truth of the method of model building
is arbitrary in the sense that
the target phenomena
doesn't directly
give us an information which method is better.
but it's not arbitrary in the sense that
I think that we can
in the same fruitful way
use the statistical mechanics,
for example,
to explain the brain
and, for example,
Bayesian modeling to explain the brain.
It's not arbitrary in the sense that
I can use the piece of paper to eat something.
Yes, I can take a glass of the water
if I want to drink,
but I cannot take a piece of paper
if I wanted to drink.
But in this sense,
when we distinct the formal principle,
the normative principle,
we distinct the model
which are built on the normative principle
and the target phenomena
which we want to explain using this model
based on the formal principle,
then it means that
the relation between the target phenomena
and the relation between the model
is not arbitrary in the sense that

we can use every kind of the model
to every kind of phenomena.
And I think that
when we try to look into our brains,
this kind of the structures
in some way motivate us
to build the computational models.
And those computational models,
I think,
is in some sense different
and the computational model,
for example,
of behavior of pigeons
or moving of my hand.
And so,
in general,
I would say that
that
truth of...
And coming back to your questions.
As I said,
the dimension of information,
the informational constraints
are more
more subtle than
the constraint like mass or temperature
because probably we don't have
as
as
as

as
as
as

as

as

doesn't don't lead us to the say something very specific ontologically independent from these models about the the nature of the phenomena we describe

thank you yeah that makes me think about a sufficient account a mechanistic account if you have a system that entirely is described by hot and cold flows through a pipe you may have a sufficient and even a total account with only thermodynamics now maybe the hot and the cold exactly signify like a signal so then there would need to be some kind of information processing model or layering to understand and and then we move into a space where

looking at just the information measures on the syntax like the amount of data transmitted in

in bytes or something over a digital connection you've already highlighted the relevance of the

semantic information flows which is um a topic that chris fields and others have started to look

into and then there's a further uncoupling between the material basis of the information transfer

and the semantic information flow

so i'll ask another question from the live chat so upcycle club wrote

are there any limitations or challenges associated with incorporating variational free energy models

into mechanisms yes good very good questions um

i think we can looking for the limitation related to the uh questions which scientist uh gives uh it means that we can looking for a limitation concerning the the the the methods and of course the more important limitation concerning the the the border of the of the systems

and um it is hard to give a one good answer because in the field of them in the field of the new mechanical philosophy there is a big discussion about the border of the mechanisms and if we take the constraint based mechanism approach the situation is looks more complicated because system traditions uh of the composing system uh the

of the composing system mechanism will claim that the border of the mechanism are related to the

border of the uh related to the components of the mechanism when i say that the components of the mechanism

is it means that something what we decompose is not enough because we have to also look into

the elements which are not literally the part of the mechanism but which but which makes the mechanism active

it means that the border of the mechanism is um unclear and i think the main uh limitation concerning my idea is to

is to say when we have to stop to explain for example

when we take an example of the pipe uh and the water moving in the pipe uh

uh this is the part of the mechanism but the the the the the the the material from which the pipe is

built is the environmental constraints which is not the part of the mechanism but which is important

which is relevant for the functioning of mechanics because the the plastic pipe if

uh makes the better for example better working of the mechanism than the wooden pipe and i think the main

limitation is related to the questions of uh which constraints for example when we talk about the

constraints as uh environmental cons uh environmental uh factors which of the constraints are

uh how many constraints we need to to say that okay this is enough we we understand how this mechanism is

collaborative um

and i don't know for now uh where this border is uh because opening the mechanism on the constraints

opening the mechanism on the information coming or processing it by the for example narrow mechanism

make these questions about the border mechanics more specific or more more complicated um but i think

this is similar discussion or similar problem like we find in the the field of the free energy principle

when we talk about the border of the mark of blankets or where we talk about the the mark of the
cognitive or mark of the living organism i think that if we integrate mechanically
the free the free energy principle with the the the for example mechanical predictive processing then
then this discussion about the border of mechanism
can be a part of the discussion of the border of uh marco blanca's of the border of uh informational
systems something like that
uh
uh
uh
thank you that's uh that's a huge point
it made me think about having a physical object could be digital or analog
that's going to implement some procedure program that is going to be bound apparently
by the halting problem like the inability to know for some kinds of systems how the pro the program is going to turn out
by the incompleteness type patterns so there's certain informational constraints that kind of come out of nowhere
seemingly to constrain
the um action
or to to provide some some
thing that limits the action or the knowability of the action so then you
um highlighted that it it is actually this stopping time question or the halting question
as a question of scientific strategy in building accounts knowing how much to include and and that's
very related to like identifying or selecting a system of interest because you could say well the pipe
comes from here and so we're going to go back up the supply chain and
pretty soon if you include the flow of air or the movement of digital information or something like
that in the system
that can unfold to include untenable models
so then there's a coarse graining
and
already um
people make informational coarse grainings like with causal influence
diagrams
like just the the the the relevance of one variable on another
and so it's just very interesting
that by having the material thermo mechanism
and a legitimated
role or status for informational
mechanisms

which are the natural kind of explanation for perhaps algorithms or information processing
that
still there's the the boundedness of what is being accounted for
but
a clearer way to know how far out informationally
just like you could go out physically in the thermal account
with layers of rooms outside of the heat bath
also you could go out
informationally more into the past for example
or you can just say we're just drawing a line around this
so maybe by
i think that
by
yeah
sorry sorry
no
i think
i think that the the the the for maybe the best way is to um
question about on the of the borders of the mechanism or borders of the system
and relativized to the our scientific interesting
because when we talk for example when we talk about the pipe
we talk about the digital system and they are some kind of artifacts and they are related to the
architecture who made them
when we talk about the brain and
it is hard to think about the brain
in the same or analogous sense and we talk about the pipe
or the glass or the paper because
there are artifacts for you can say about natural artifacts and i think that
our
thinking about the borders of the system we want to describe
maybe primary should be related to the our
our questions concerning the borders
and prob maybe it the story was that that
mechanisms
and
the system
and
in this system tradition the traditional viable mechanism

and
will say something like that
we think about the brains as
mechanism like the other one
we can we can develop the the structure of the the of the social web the structure of the
the the of the brain the structure of the the bones and so on and so on
and this this way of thinking about the brain as the kind of the mechanics like the other one
in some sense
direct the investigation to brains and this is why for example decomposition
as the whole strategy in new mechanical philosophy many others
we think about this as as relevant and uh we don't need anything else
but when bechdel for example claims that okay but there is a mistake because our brains
are based on the web of active mechanism and those active mechanism
is most more more specific because uh what makes them active is not only
it's not it is not only what makes them uh and mechanism in traditional normal sense
and when we talk when we look into the paper of bechdel
uh um he doesn't develop the the idea of talking about the whole brain they developed the investigation
into the small pieces of how some kinds of cells transform uh molecules and so on and so on but
they they still claims that if we uh show the bound in two
too fast then uh then we don't explain
correctly how the given mechanism the given cells uh do actively what that's what actively what it does
and um i don't know if if um probably if i want to the defend the idea of moderate realism
about bayesian modeling i would like to say something like that that
well that there are some kind of borders in the world and we try to looking for those borders
and maybe this the systems and the difference between different systems uh motivate us to
uh trying to describe these borders but uh there is no possibility to say that
the factor here there is a borders but always when we talk about the borders this is talking about
the borders is related to our uh our measurement tools and the methods so i think that um the small
move in mechanical philosophy which i describe as the constraint-based mechanics approach um we can
interpretate as um as a example of the situation when the phenomena motivate us to change our scientific
practice because uh back 12 20 or 30 30 years developed the new mechanical philosophy but
in the last five or six years he claims okay but this old view was not perfect we have to change
something and i think this is this is the argument for the moderate interpretation of the scientific
practice phenomena in some sense because we developed a good method investigating phenomena phenomena in
some sense motivate us to change to do to uh develop the new methods and change our our scientific uh heuristic and so on so
wow that's awesome one other thing it made me think about was the differences the the relevance or the

the difference between the thermodynamic and the informational accounts become very clear when studying decentralized systems that are spatially disaggregated because when there's something that's physically enclosed and and it has like a unitary physical component and a unitary informational component like a desktop computer or a single tissue then there is an undeniable um overlap between those two systems which kind of host and relate to each other but then when there's a system that um has other things amidst it that then just thinking here about nest mates and the ant colony well then spatial enclosure and just inclusion of all the material components

seems to clearly capture too much and so

so that seems to suggest a more informational role which takes on many rich areas from semiotics and all these other semantic questions

not just again a reduction of the informational to the syntactic informational like how much

um energy or material was transferred because again that will just

conflate that'll make mechanism it's it's it's um it's a approach that won't lead to the best research agenda

on mechanistic accounts

interesting

um

But I think what is important here is what Tristan's really important claims, that I think that

information changes everything in that sense that when you try to literally think about

Bayesian approximation and so on, then it means that everything we want to describe and explain as

scientists is strictly related to the information we receive. And of course this is very simple, but

every object we observe, every object we want to describe

is always related to the information we have about the subject. And it means that there is no

something like we've got a naked fact, naked state of affairs.

Maybe ontologically there are, but from the point of the view of the finite entities like us,

every time when we receive some information, every time when we observe something and we try to develop

a model of some kind of phenomena, it's always related to some kind of information, what we have about

this phenomena. And I try to say something very metaphysical.

I think that the variational

point of the view, as we can say something like that, the Bayesian point of the view shows us that there is no

distinction border between the object we have to map between the language we try to use to map this object.

Because I think there is something that Maxwell Rammstadt has right when we claim that

which still we try to describe the objects who behave like the...

that we try to describe the territory which behave like a map, which describes this territory.

And of course this is not only the metaphor, I think that

everything what scientists try to do is to reconfigure, is to identify

the structures which are extended now to the structure of thinking, structure of information,

which we use. I don't know.

I think that that Bayesian
view about human brains, about the world, lead us to the one very important question about
the informational nature of the world. I don't know because...
I speak right now. Okay, but when we think, when we take seriously this kind of thinking about the
science, about the entities, the most basic, the most fundamental future of the world is not a mass,
it's not a temperature, it's not an energy, but an information. And I think the information is
something that could change a difference. And literally it means that if we want to
develop a science, we cannot ignore this very basic, fundamental futures of the world.
Why? Because the information... of course, I don't want to say it's something like panpsychism,
non-informational, something like that. But I will say that information is always for someone.
And I would like to say that...
from some time... when I was reading the papers of Ristam and so on, I still think about why we should or
why we shouldn't ignore the information in our investigation. And I think that Bayesian modeling
give us a good tool maybe is a plea for the place of information in the science.
Because...
But not only literally talking about information, okay, we know, it's obvious that brains process
information, but information as entity which making possible many, many of the phenomena. And I think that
our brains... our brains... in short, we say, our brains are the entities which are...
which are possible because there are information which makes this mechanism, this kind of mechanism,
but not the other. Sorry, I think this is really speculative, metaphysical. But it's hard to
catch this idea.
Yeah, there's a lot there. Certainly, the unity of the scientific and the broader informational
endeavor is very interesting. And so thinking about that explicitly in terms of epistemic foraging,
seeking out which measurements or observations to accrue. And that being kind of on a continuum of agency of
experimental selection. And there's other... Like a fixed camera does not have control over the sensory
observations it's making it could have internal cognitive control but but having a unified way to
talk about what it is that science or just different ways of knowing even outside of what
we might call professional science today that those different manifestations are are part of a
ecosystem of shared intelligence part of a shared fabric with other information processing
as chemical elements are in circulation in a chemical ecosystem
what is the status of the semantic informational
connections in semantic ecosystems and people could have viewpoints like well there are no
semantic ecosystems there's just syntactic ones and you know there's no syntactic ones
there's just material ones i mean people can always go that reductionist path
but to the point where any given person says okay i've kind of bottomed out on reductionism
here or i'm looking for another perspective there is going to be the question of what the semantic
account is and what its status is and does it articulate with the thermodynamic account

do they have a a zone of indistinguishability do they have um a broadly overlapping

region maybe it could be one way in one system in a different way in a different system but these

will constitute like some of the fundamental philosophy questions that that you identified in terms of

real mechanistic accounts for intelligent systems and using active inference to develop any kind of account

you know

um

i think we cannot ignore the semantic uh and

and if

maybe the one of the most important things that fristom says is that semantic

is important in that sense that

it is not only the medium uh which we use for example when we

if i correctly understand the the origins of the the the frestonian free energy principle it comes from the method of measurement of the brain activity

and

uh

uh

someone can say okay this is only the method and there is no any implication uh

of course but i think that the fristom will

say something more it says that

there are phenomena which in which semantics play and not only the uh

uh

of course there is phenomena in which semantics play an important role

uh

uh

but

we cannot ignore the semantic dimension

uh

in our practice of uh science building or model building

uh it means that

uh

semantic is not something like the piece of the paper i can use the piece of the paper when i wrote a paper or i can

uh use a word or offix and something like that and this is not important what a material i use

i think that many authors or maybe many papers think about the semantic as some kind of the medium like the

kind of the paper i use

i think that the uh

uh

free energy relative inference framework show us that

the information or semantic is something

uh

what makes possible

uh

for example to to to to to do this to do a science or make it or maybe the information semantic is some is something what makes possible

uh

for us a human human being

and

maybe like a thermodynamic

free energy

with if we don't uh

transform the the molecules from the environment we don't live but

the question is why uh

the question is why uh

uh science

uh talk

uh many things about

the uh thermodynamic

processes

and talk enough not enough

uh about the informational processes

i think that the bayesian smodeling uh free stone approach lead us to the conclusion that

there are systems for which the transfer of information is uh

uh

uh

uh

important like to transform process the thermodynamic energy in the argument which i tried to develop this from the neural computation so as that

always there is a trade of there is a compromise there is a relation between information processing and thermodynamic work caused if there is some kind of trade of why we don't take and use it but probably it as i said information is something what is hard to be measured like temperature right

is some kind of trade-off why we don't take and use it. But probably, as I said, information is

something that is hard to be measured, like temperature, like mass, and so on. And I believe

that rational principles give us a chance to try to develop this. Of course, remembering about this, that those principles are only the, those principles lead us to the tools which are only the

more or better, more or minor, the approximation of the real phenomena.

Okay. Well, wow. Do you have any closing thoughts or where is your research in this area going to go?

I think that I've got a, I wrote a grant proposal about this idea. And a few days ago, I received a reject in Polish

center. And the one of them, I think this, the only one, the reason why they reject my proposal is that I

I don't propose using or apply this idea into the real modeling framework. So the next steps, I, I, and I think I agree,

because this is very speculative. I propose some kind of heuristics. And I believe that this heuristic can be

applied in a real model building strategy. And the next step I would now I'm thinking is that to meet person or to

make a competence concerning that and find the, finding the way to how to taste this idea in practice.

And still there is two options. And the first one is that, okay, this is the speculative idea and we don't need

to think about the mechanism in terms of information constraints and so on and so on. And the second is that, okay, this is a

correct idea, but, but maybe a variational principle by Asian modeling is useless here because

everything we need is coming from the thermodynamic of information or the classical theory of information, like, and so on.

So I think there are all many open doors. And I hope I will take away to find some of them.

Thank you. Well, it was a great presentation and discussion. You're welcome back anytime to share

the next step. I really appreciate this. I think it certainly is a great benefit to the literature to

connect the kinds of things that people want to say and do to now how we are saying and doing it.

Thank you. Thank you, Daniel. Thank you very much. Thanks for questions.

Thank you. That was a very nice experience. Thank you. Excellent. All right. Till next time.

Good night. Bye. Bye.

Bye.

Thank you.